Logarithmus

$b^x = a$ Exponent x gesucht

Basis $b > 0$ und $a > 0$ gegeben

Der Wert von x wird als Logarithmus von a zur Basis b bezeichnet.

$$x = \log_b(a)$$

Bsp. $\log_2(16)$ "Zwei hoch wie viel ist 16?"

bzw. "Wie lässt sich 16 als 2er-Potenz schreiben?"

$$2^x = 16 \implies x = 4$$

$$\Rightarrow \log_2(16) = 4$$

$$\log_4\left(\frac{1}{16}\right) = \log_4\left(\frac{1}{4^2}\right) = \log_4\left(4^{-2}\right) = -2$$

$$\log_{125}(5) = \log_{125}\left(\sqrt[3]{125}\right) = \log_{125}\left(125^{1/3}\right) = \frac{1}{3}$$

$125^x = 5$, $\sqrt[3]{125} = 5$, $125 = 5^3$

$x = ?$

Ü5.1 a) $\log_2(32) = ?$ "2 hoch wie viel ist 32 ?"

$\log_2(32) = x$

$\Longleftrightarrow \quad 2^x = 32$

$\log_2(32) = 5$

b) $\log_{10}(0.1) = ?$

$\log_{10}(0.1) = \log_{10}\left(\frac{1}{10}\right) = \log_{10}(10^{-1}) = -1$

c) $\log_5(25) = 2$

d) $\log_6\left(\frac{1}{36}\right) = \log_6\left(\frac{1}{6^2}\right) = \log_6(6^{-2}) = -2$

e) $\log_4(1) = 0$, $\quad 4^0 = 1$

f) $\log_{10}(\sqrt[3]{100}) = x \Longleftrightarrow 10^x = \sqrt[3]{100}$

$= \sqrt[3]{10^2}$

$= 10^{2/3}$

$\Rightarrow \log_{10}(\sqrt[3]{100}) = \frac{2}{3}$

g) $\log_2(0.125) = \log_2\left(\frac{1}{8}\right) = \log_2\left(\frac{1}{2^3}\right)$

$= \log_2(2^{-3}) = -3$

h) $\log_9\left(\frac{1}{3}\right) = ?$ "9 hoch was ist $\frac{1}{3}$?"

$9^x = \frac{1}{3}$, $\quad 9^{1/2} = \sqrt{9} = 3$

$$9^{-1/2} = \frac{1}{3} \quad \Leftarrow$$

$$\Rightarrow \log_9\left(\frac{1}{3}\right) = -\frac{1}{2}$$

i) $\log_3(9^9) = \log_3\left((3^2)^9\right) = \log_3(3^{18}) = 18$

j) $\log_2(\sqrt[5]{16}) = \log_2\left(\sqrt[5]{2^4}\right) = \log_2(2^{4/5}) = \frac{4}{5}$

k) $\log_{16}(2) = x \Leftrightarrow 16^x = 2$

$16 = 2^4$ also $16^{1/4} = 2$

$$\Rightarrow \log_{16}(2) = \frac{1}{4}$$

l) $\log_{27}\left(\frac{1}{81}\right) = x \Leftrightarrow 27^x = \frac{1}{81}$

$$(3^3)^x = \frac{1}{3^4}$$

$$3^{3x} = 3^{-4}$$

$$x = -\frac{4}{3}$$

<u>Oder:</u> $\log_{27}\left(\frac{1}{81}\right) = \log_{27}\left(\frac{1}{3^4}\right) = \log_{27}\left(\frac{1}{\sqrt[3]{27}^4}\right)$

$$= \log_{27}\left(\frac{1}{27^{4/3}}\right) = \log_{27}\left(27^{-4/3}\right) = -\frac{4}{3}$$

$$\Rightarrow \log_{27}\left(\frac{1}{81}\right) = -\frac{4}{3}$$

m) $\log_{\frac{1}{4}}(\sqrt{8}) = \log_{\frac{1}{4}}(\sqrt{2^3}) = \log_{\frac{1}{4}}(2^{3/2})$

$= \log_{\frac{1}{4}}\left((4^{1/2})^{3/2}\right) = \log_{\frac{1}{4}}(4^{3/4})$

$= \log_{\frac{1}{4}}\left(\left(\left(\frac{1}{4}\right)^{-1}\right)^{3/4}\right) = \log_{\frac{1}{4}}\left(\left(\frac{1}{4}\right)^{-\frac{3}{4}}\right) = -\frac{3}{4}$

Ü 5.2

a) $\log_2(3) = x$, $2^x = 3$

Zwischen welchen ganzen Zahlen liegt x?

$2^1 = 2$, $2^2 = 4$

$\Rightarrow \quad 1 < \log_2(3) < 2$

b) $\log_2(30) = ?$ $\quad 2^4 = 16 < 30 < 2^5 = 32$

$\Rightarrow \quad 4 < \log_2(30) < 5$

c) $\log_{10}(0.07) = ?$ $\quad 10^n < 0.07 < 10^m$

$0.01 < 0.07 < 0.1$

$10^{-2} < 0.07 < 10^{-1}$

$\Rightarrow \quad -2 < \log_{10}(0.07) < -1$

d) $\log_8(2^{17}) = ?$

$$\underset{\text{exakt möglich}}{=} \log_8\left(\sqrt[3]{8}^{\,17}\right) = \log_8\left(8^{17/3}\right) = \frac{17}{3}$$

e) $\log_{0.5}(18)$

$$2^4 < 18 < 2^5$$

$0.5 = 2^{-1}$ $\qquad \left(\left(\frac{1}{2}\right)^{-1}\right)^4 < 18 < \left(\left(\frac{1}{2}\right)^{-1}\right)^5$

$\left(\frac{1}{2}\right)^{-1} = 2$ $\qquad \left(\frac{1}{2}\right)^{-4} < 18 < \left(\frac{1}{2}\right)^{-5}$

$$-5 < \log_{0.5}(18) < -4$$

Einfache Exponentialgleichungen lösen

<u>Bsp.</u> a) $6^x = 12 \iff x = \log_6(12) \overset{TR}{=} 1.387$

Mit TR evtl. $x = \dfrac{\log_{10}(12)}{\log_{10}(6)} = \dfrac{\ln(12)}{\ln(6)}$

b) $\dfrac{4^x}{5^x} = 7 \iff \left(\dfrac{4}{5}\right)^x = 7$

Ü 5.3 $\qquad \iff x = \log_{\frac{4}{5}}(7) = -8.72$

a) $20 = 4^x \iff x = \log_4(20) \approx 2.161$

b) $\frac{2}{3^x} = 9 \iff \frac{2}{9} = 3^x \iff x = \log_3\left(\frac{2}{9}\right) = -1.369$

c) $8^x = 10 \cdot 5^x \iff \left(\frac{8}{5}\right)^x = 10 \iff x = \log_{\frac{8}{5}}(10) = 4.899$

d) $x^2 = 10 \iff x = \pm\sqrt{10} = \pm 3.162$

e) $100 = 2^{x+1} \iff x+1 = \log_2(100)$

$x = \log_2(100) - 1 = 5.644$

<u>oder:</u> $100 = 2^x \cdot 2 \iff 50 = 2^x$

$\iff x = \log_2(50)$

f) $2^{-x} = 9 \iff -x = \log_2(9)$

$\iff x = -\log_2(9) = -3.170$

g) $40 = \frac{5}{4^x} \iff 4^x = \frac{1}{8} \iff x = \log_4\left(\frac{1}{8}\right)$

$x = \log_4\left(\frac{1}{8}\right) = \log_4\left(\frac{1}{2^3}\right) = \log_4\left(\frac{1}{(4^{1/2})^3}\right) = -\frac{3}{2}$

h) $\frac{20}{x^3} = 7 \iff \frac{20}{7} = x^3 \iff x = \sqrt[3]{\frac{20}{7}}$

i) $10 = 0.5^{3-x} \iff 3 - x = \log_{0.5}(10)$

$x = 3 - \log_{0.5}(10) = 6.322$

Logarithmengesetze

$$\log_2(x \cdot y) = \log_2(2^u \cdot 2^v), \quad \text{mit } u = \log_2(x),\ v = \log_2(y)$$

$$= \log_2(2^{u+v})$$

$$= u + v = \log_2(x) + \log_2(y)$$

analog: $\log_2\left(\frac{x}{y}\right) = \log_2(x) - \log_2(y)$

$$\log_2(x^y) = \log_2\left((2^u)^y\right) = \log_2(2^{u \cdot y})$$

$$= u \cdot y = y \cdot \log_2(x)$$

Logarithmengesetze mit allg. Basis a> 0

(L1) $\log_a(x \cdot y) = \log_a(x) + \log_a(y)$

(L2) $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$

(L3) $\log_a(x^y) = y \cdot \log_a(x)$

Ü 5.4

a) $\log_a(x^2 \cdot y^2) \overset{(L1)}{=} \log_a(x^2) + \log_a(y^2)$

$\overset{(L3)}{=} 2 \cdot \log_a(x) + 2 \cdot \log_a(y)$

b) $\log_a(x \cdot \sqrt{y}) = \log_a(x) + \log_a(\sqrt{y})$

$= \log_a(x) + \log_a(y^{1/2})$

$= \log_a(x) + \frac{1}{2} \log_a(y)$

c) $\log_a(\sqrt[3]{x^2 y z^3}) = \log_a((x^2 y z^3)^{1/3})$

$= \frac{1}{3} \log_a(x^2 y z^3) = \frac{1}{3}(2\log_a(x) + \log_a(y) + 3\log_a(z))$

d) $\log_a(\sqrt{x \cdot \sqrt{y}}) = \log_a((x \cdot y^{1/2})^{1/2})$

$= \frac{1}{2}(\log_a(x) + \frac{1}{2}\log_a(y))$

e) $\log_a\left(\frac{1}{x^2 y^3}\right) = \log_a(x^{-2} \cdot y^{-3})$

$= -2\log_a(x) - 3\log_a(y)$

f) $\log_a\left(\sqrt[5]{\frac{x^2}{y^3}}\right) = \frac{1}{5}\log_a\left(\frac{x^2}{y^3}\right) = \frac{1}{5}(\log_a(x^2) - \log_a(y^3))$

$= \frac{1}{5}(2\log_a(x) - 3\log_a(y))$

g) $\log_6\left(\frac{36(x+y)}{x^6 \cdot \sqrt[6]{y}}\right) = \log_6(36(x+y)) - \log_6(x^6 \cdot y^{1/6})$

$= \log_6(36) + \log_6(x+y) - \left(6\log_6(x) + \frac{1}{6}\log_6(y)\right)$

$= 2 + \underbrace{\log_6(x+y)}_{\text{kann nicht weiter umgeformt werden!}} - 6\log_6(x) - \frac{1}{6}\log_6(y)$

h) $\log_5\left(\frac{x^4 \cdot y^5}{625}\right) = \log_5(x^4) + \log_5(y^5) - \log_5(625)$

$= 4 \cdot \log_5(x) + 5 \cdot \log_5(y) - 4$

Ü 5.5 a) $\log_a\left(\frac{2}{3}\right) + \log_a\left(\frac{3}{4}\right) + \log_a\left(\frac{4}{5}\right) = \log_a\left(\frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5}\right)$

$= \log_a\left(\frac{2}{5}\right)$

b) $2\log_a(3) + \frac{1}{3}\log_a(4) - 4\log_a(2)$

$= \log_a(3^2) + \log_a(4^{1/3}) - \log_a(2^4)$

$= \log_a\left(\frac{3^2 \cdot 4^{1/3}}{2^4}\right) = \log_a\left(\frac{3^2 \cdot 2^{2/3}}{2^4}\right)$

$= \log_a\left(3^2 \cdot 2^{-\frac{10}{3}}\right) = \log_a\left(\frac{3^2}{\sqrt[3]{2^{10}}}\right)$

c) $\log_a(x) - \log_a(\sqrt{x}) = \log_a\left(\frac{x}{\sqrt{x}}\right) = \log_a(\sqrt{x})$

$\frac{x}{\sqrt{x}} = \frac{x}{x^{1/2}} = x^{1-\frac{1}{2}} = x^{1/2}$

d) $\log_a(2x) - 2\log_a(x) = \log_a(2x) - \log_a(x^2)$

$= \log_a\left(\frac{2x}{x^2}\right) = \log_a\left(\frac{2}{x}\right)$

e) $(n+1)\log_a(x) - \frac{1}{3}\log_a(x^{6n})$

$= \log_a(x^{n+1}) - \log_a\left((x^{6n})^{1/3}\right)$

$= \log_a(x^{n+1}) - \log_a(x^{2n})$

$= \log_a\left(\frac{x^{n+1}}{x^{2n}}\right) = \log_a(x^{-n+1})$

f) $1 + \log_2(9) = \log_2(2) + \log_2(9) = \log_2(18)$

g) $\log_{10}(400) - 2 = \log_{10}(400) - \log_{10}(100) = \log_{10}(4)$

h) $\frac{\log_a(x+y) + \log_a(x-y)}{2} - \log_a(x^2-y^2)$

$= \frac{1}{2}\left(\log_a((x+y)\cdot(x-y))\right) - \log_a(x^2-y^2)$

$= \frac{1}{2}\log_a(x^2-y^2) - \log_a(x^2-y^2)$

$= -\frac{1}{2}\log_a(x^2-y^2) = \log_a\left((x^2-y^2)^{-1/2}\right) = \log_a\left(\frac{1}{\sqrt{x^2-y^2}}\right)$